

TIMO Primary 1 Practice Set

75 English Practice Questions

Logical Thinking - Arithmetic - Number Theory - Geometry - Combinatorics

Name: _____
Date: _____
Score: _____ / 75

Instructions: Read each question carefully. Show your thinking when needed. Some questions have more than one step.

Section A - Logical Thinking

Balance Problems

1. A red ball weighs the same as 3 blue balls. How many blue balls have the same weight as 2 red balls? _____
2. Two apples balance 8 small oranges. How many small oranges balance 1 apple? _____
3. A box and 2 blocks balance 7 blocks. How many blocks have the same weight as the box? _____

Basic Number Pattern & Sequence

4. Find the missing number: 2, 5, 8, 11, __, 17.
5. What comes next? 20, 18, 16, 14, __.
6. Find the missing number: 1, 3, 6, 10, __.

Basic Figure Pattern

7. A shape pattern is: circle, triangle, square, circle, triangle, square, ... What is the 11th shape?
8. A pattern is: star, star, circle, star, star, circle, ... What is the 14th shape?
9. A pattern repeats: triangle, circle, circle. How many triangles are there in the first 12 shapes?

IQ Age Problem & Date Problem

10. Mia is 6 years old. Ben is 3 years older than Mia. How old is Ben? _____
11. Tom is 8 years old now. How old will he be 4 years from now? _____
12. Today is Monday. What day will it be 5 days later? _____

Guess on 2-digit Numbers

13. I am a 2-digit number. My tens digit is 4 and my ones digit is 3. What number am I?
14. I am greater than 50 but less than 60. My ones digit is 7. What number am I?
15. My digits add to 9. My tens digit is 2. What number am I?

Section B - Arithmetic

Smart Addition on 1-digit Numbers with Carrying

16. Find a smart way to add: $8 + 7 + 2 + 3 =$ _____
17. Calculate: $9 + 6 + 1 + 4 =$ _____
18. Calculate: $7 + 8 + 3 + 2 + 5 =$ _____
19. Nina has 8 red beads and 7 blue beads. She gets 5 more beads. How many beads does she have now?

Smart Subtraction on 1 to 2-digit Numbers with Carrying

20. Calculate: $32 - 18 =$ _____
21. Calculate: $50 - 27 =$ _____
22. Sam has 41 stickers and gives away 16. How many stickers are left?
23. Which is greater: $45 - 19$ or $40 - 13$? Write the greater answer.

Multiplication on 1 to 2-digit Numbers without Carrying

24. There are 4 bags with 6 marbles in each bag. How many marbles are there?
25. Calculate: $12 \times 3 =$ _____
26. A spider has 8 legs. How many legs do 5 spiders have?
27. There are 11 pencils in each box. How many pencils are in 4 boxes?

Balance on an Equation

28. Fill in the box: $8 + [] = 5 + 9$.
29. Fill in the box: $17 - [] = 6 + 4$.
30. Fill in the box: $[] + 7 = 9 + 8$.

Section C - Number Theory

Introduction to Odd & Even

31. Which number is even: 13, 18, 21, or 25?
32. Is $7 + 9$ odd or even?
33. Three consecutive numbers are 10, 11, 12. How many of them are even?

Mathematical Leveling

34. Put the correct sign $>$, $<$, or $=$: $8 + 7$ ____ $10 + 5$.
35. Put the correct sign $>$, $<$, or $=$: 4×5 ____ 18.
36. Which is smallest: $9 + 8$, $20 - 4$, or 3×6 ?

Advanced Fibonacci Series

37. Find the next two numbers: 1, 1, 2, 3, 5, 8, __, __.
38. Find the missing number: 2, 3, 5, 8, 13, __.
39. A sequence starts 1, 2. Each new number is the sum of the previous two. Write the next four numbers.

Match Equation

40. Which has the same answer as $7 + 8$? A) $10 + 4$ B) $9 + 6$ C) $8 + 8$ D) $6 + 7$
41. Which has the same answer as $20 - 7$? A) $6 + 7$ B) $8 + 6$ C) $15 - 1$ D) 3×5
42. Complete: $6 \times 4 = 20 +$ _____.

Basic Arithmetic Pattern

43. Find the missing number: 3, 6, 9, 12, __.
44. Find the missing number: 5, 10, 20, 40, __.
45. Look at the rule: $2 \rightarrow 6$, $3 \rightarrow 9$, $4 \rightarrow 12$. Then $7 \rightarrow$ _____.

Section D - Geometry

Counting 2-D & 3-D Figures

46. A large square is divided into 4 equal small squares. How many squares are there altogether, including the large square?
47. Two separate triangles and three separate circles are shown. How many 2-D shapes are there altogether?
48. You have 3 cubes and 2 balls. How many 3-D objects are there altogether?
49. A tower is made from 2 cubes on the bottom and 1 cube on top. How many cubes are used?

Counting Sides & Interior Angles

50. How many sides does a pentagon have?
51. How many corners does a rectangle have?
52. Two triangles are separate. How many sides do they have altogether?
53. A square and a triangle are separate. How many corners do they have altogether?

Distinction of 2-D Figures

54. Which shape has no straight sides: circle, triangle, square, or rectangle?
55. Which shape has exactly 3 sides?
56. Which two shapes always have 4 sides: square, circle, rectangle, triangle?
57. I have 4 equal sides and 4 corners. What shape am I?

Basic Figure Pattern

58. The pattern is: square, circle, triangle, square, circle, triangle, ... What is the 8th shape?
59. The pattern is: circle, square, square, circle, square, square, ... What is the 10th shape?
60. Row 1 has 1 triangle, Row 2 has 2 triangles, Row 3 has 3 triangles. How many triangles should Row 5 have?

Section E - Combinatorics

Seven Bridges of Konigsberg - Path Thinking

- 61.** A path has three points A-B-C connected in a line. Starting at A, can you cross both bridges exactly once and finish at C?
- 62.** Three islands A, B, C have bridges A-B, B-C, and A-C. Starting at A, can you cross every bridge exactly once and return to A?
- 63.** A path is A-B-C-D. How many bridges must you cross to go from A to D?

Arranging Numbers in Orders

- 64.** Arrange 7, 2, 9, 4 from smallest to largest.
- 65.** Arrange 15, 11, 19, 13 from largest to smallest.
- 66.** Using 2, 5, 8 once each, make the largest 3-digit number.

Simple Distribution

- 67.** 8 candies are shared equally between 2 children. How many candies does each child get?
- 68.** 12 pencils are shared equally among 3 students. How many pencils does each student get?
- 69.** 15 cookies are placed equally on 5 plates. How many cookies are on each plate?

Counting on Specific Numbers

- 70.** How many times does digit 2 appear in: 2, 12, 20, 21, 22?
- 71.** How many numbers from 1 to 10 are even?
- 72.** How many numbers in 3, 6, 9, 12, 15, 18 are greater than 10?

Formation of a 3-digit Number

- 73.** Use 1, 3, and 5 once each. Make the largest 3-digit number.
- 74.** Use 2, 4, and 7 once each. Make the smallest 3-digit number.
- 75.** Using 1, 2, and 3 once each, how many different 3-digit numbers can you make?

Answer Key

1. 6	2. 4	3. 5
4. 14	5. 12	6. 15
7. triangle	8. star	9. 4
10. 9	11. 12	12. Saturday
13. 43	14. 57	15. 27

16. 20	17. 20	18. 25
19. 20	20. 14	21. 23
22. 25	23. 27	24. 24
25. 36	26. 40	27. 44
28. 6	29. 7	30. 10

31. 18	32. even	33. 2
34. =	35. >	36. 20 - 4 (16)
37. 13, 21	38. 21	39. 3, 5, 8, 13
40. B) 9 + 6	41. A) 6 + 7	42. 4
43. 15	44. 80	45. 21

46. 5	47. 5	48. 5
49. 3	50. 5	51. 4
52. 6	53. 7	54. circle
55. triangle	56. square and rectangle	57. square
58. circle	59. circle	60. 5

61. Yes	62. Yes	63. 3
64. 2, 4, 7, 9	65. 19, 15, 13, 11	66. 852
67. 4	68. 4	69. 3
70. 6	71. 5	72. 3
73. 531	74. 247	75. 6

Teacher note: Questions are arranged to cover all topics in the provided TIMO Primary 1 outline. Several items intentionally require pattern recognition or more than one mental step.